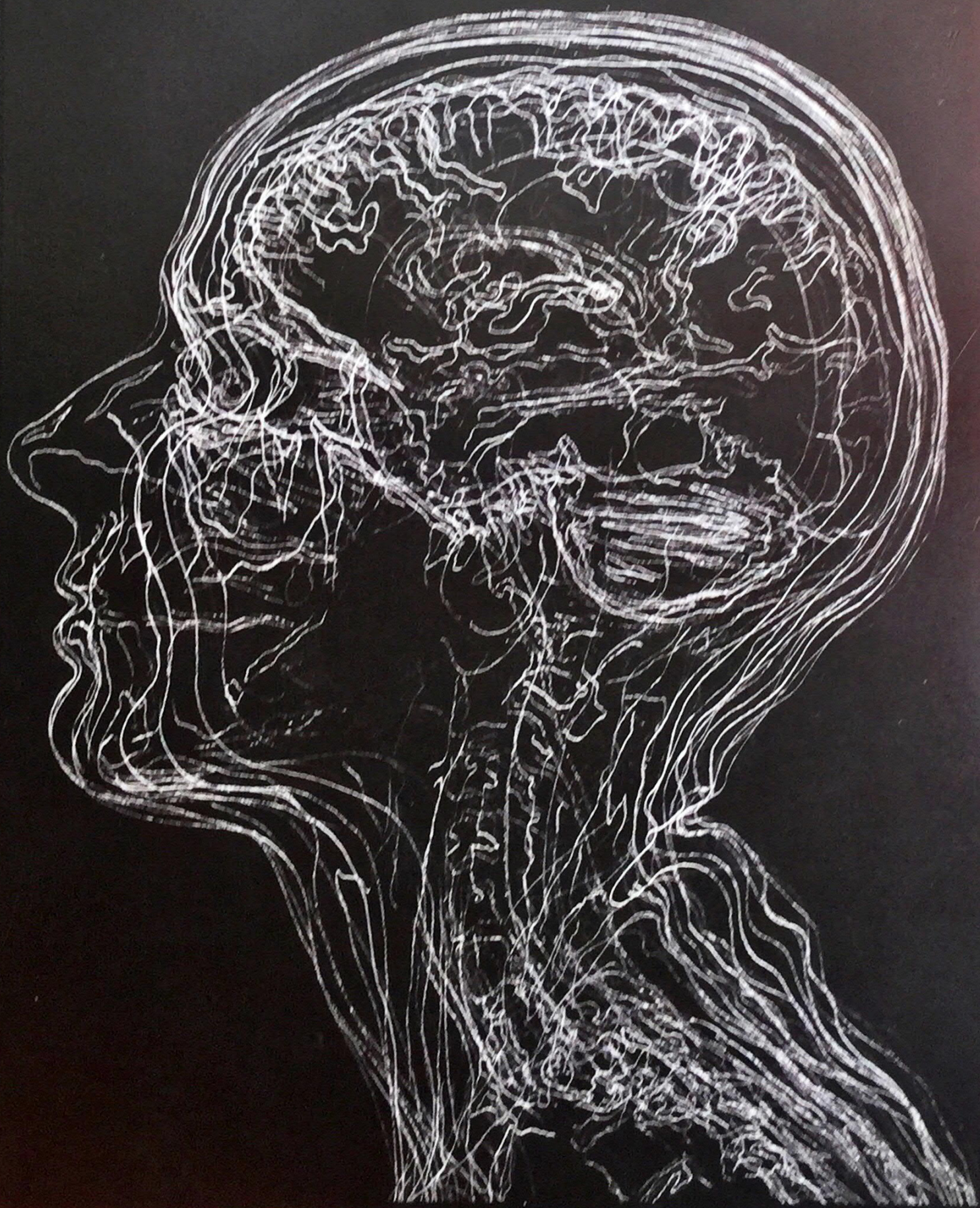


FRCS (Neurosurgery) Viva Case Book

Mr Ian Anderson and Professor Paul Marks



Cover Art:

"Self Portrait" by Angela Palmer

Photograph courtesy of Richard Holtum

Copyright © Angela Palmer. All Rights Reserved.

Kindly reproduced with permission from the artist & photographer.

Website: <http://www.angelaspalmer.com>

FNS

Publications

Suite 268, 33 Great George Street, Leeds LS1 3AJ

FNS Publications is the publishing division of Forensic Neurosciences Limited

FNS Publications is a registered trademark in the UK

Published in the United Kingdom by FNS Publications

© Ian Anderson & Paul Marks

First Published in 2013

ISBN 978-0-9566595-2-1

All rights are reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, without the prior permission in writing of FNS Publications, or as expressly permitted by law, or under terms agreed with the appropriate reprographic rights organization. Enquiries concerning reproduction outside the scope of the above should be sent to the Rights Department, FNS Publications, at the above address.

You must not circulate this book in any other binding or cover and you must not impose the same condition on any acquirer.

Printed in Great Britain on acid-free paper by Lavenham Press Ltd.

Contents:

Foreword *by Dr Robert Spetzler M.D.*

Introduction

Section 1: Basic & Allied Sciences Vivas 1-12

Section 2: Cranial Neurosurgery Vivas 13-23

Section 3: Diagrams Vivas 24-28

Section 4: Paediatric Neurosurgery Vivas 29-40

Section 5: Peripheral, Functional & Neurology Vivas 41-47

Section 6: Spinal Neurosurgery Vivas 48-63

Section 7: Vascular Neurosurgery Vivas 64-68

Section 8: Medico-Legal Considerations Vivas 69-72

Appendix

Foreword:

By Dr Robert Spetzler M.D.

J.N. Harber Chairman of Neurological Surgery and Director, Barrow Neurological Institute,
Phoenix, Arizona.

Professor of Surgery, Section of Neurosurgery, University of Arizona College of Medicine,
Tucson, Arizona.

This volume is a thoughtful compendium of knowledge to provide answers to commonly encountered board questions allowing the candidate to prepare appropriately for this always stressful rite of passage in our specialty.

Drs Marks and Anderson are to be congratulated for this most worthwhile addition to the neurosurgical literature; it will be welcomed by those preparing for their board ordeal and will be a great refresher for those that already are board certified.

Dr Robert Spetzler

Introduction:

This book is intended as a guide to the clinical phase of the exit exams in neurosurgery. While principally aimed at the FRCS qualification, it could as easily be applied to the equivalent board reviews in other countries. Rather than didactic text book, this guide is designed to aid learning through real clinical cases and problem-solving. There is a mixture of core, "Bread and butter" neurosurgery and rarer conditions to keep the reader interested and to differentiate between levels of knowledge (as in the real examination). It is hoped that this text will afford readers the opportunity to "Learn through doing"; and there is no better way to progress than this!

For each viva, the questions are on the odd (right-sided) page, with the answers given overleaf.

Huge thanks must go to each and every neurosurgical consultant at Leeds General Infirmary, who between them, have taught me far more about neurosurgery than I could ever have learned from reading alone.

We would also like to thank: Louise Dunsmure, Azzam Ismail, Tufail Patankar and Emma Owen for their help with some of the cases in this book.

Mr Ian Anderson

MB ChB, BSc (Hons), MRCS(Ed)

Specialist Neurosurgical Registrar, Leeds General Infirmary

I have examined for the Intercollegiate Fellowship in Neurosurgery for several years and am an external examiner for comparable exit examinations overseas. Candidates who have prepared well, who have done the job and read around the subject tend to have little problem with passing whereas those who have been complacent or have not read widely are readily obvious to the examiners and are likely to fail.

The FRCS (Neurosurgery) is a challenging and searching examination but with proper preparation and the development of good examination technique it is not an insuperable hurdle.

This book should help you with preparation and give you an idea of the sort of questions you may encounter in the oral part of the examination. As with all skills and examinations are no exception, performance will improve with practice. You should remember the old adage of the British Army of the 7 "Ps" (proper planning and preparation prevents piss poor performance)!

Professor Paul Marks

BA,LLM,MD,FRCS,FFFLM,FCS(ECSA)

Consultant Neurosurgeon Emeritus, Leeds General Infirmary

Honorary Consultant Neurosurgeon, Harrogate District Hospital

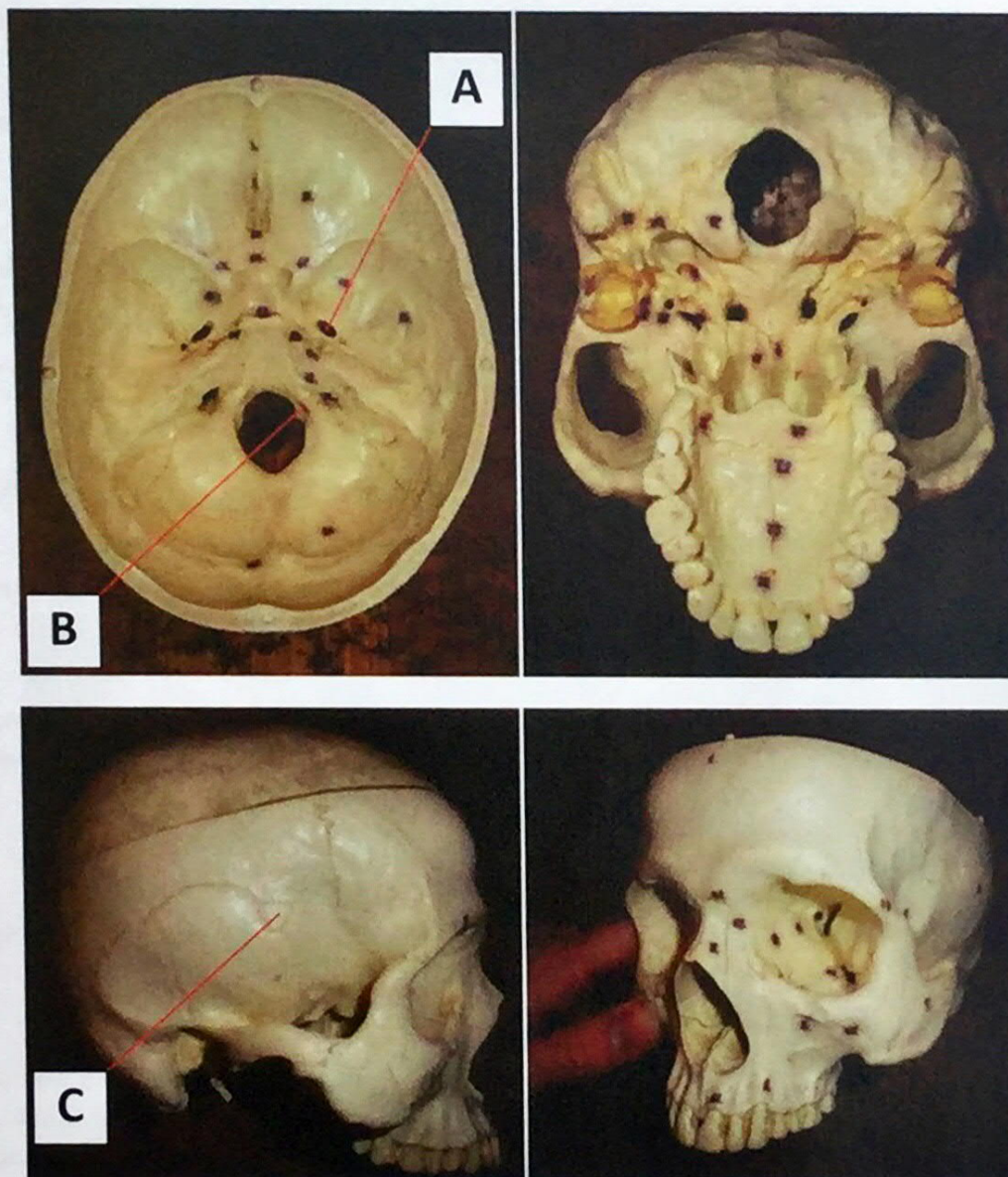
H.M. Coroner for Hull and the East Riding of Yorkshire

Visiting Professorial Fellow in Law, St Chad's College, University of Durham

Vice President of the Faculty of Forensic and Legal Medicine

Section 1:

Basic & Allied Sciences

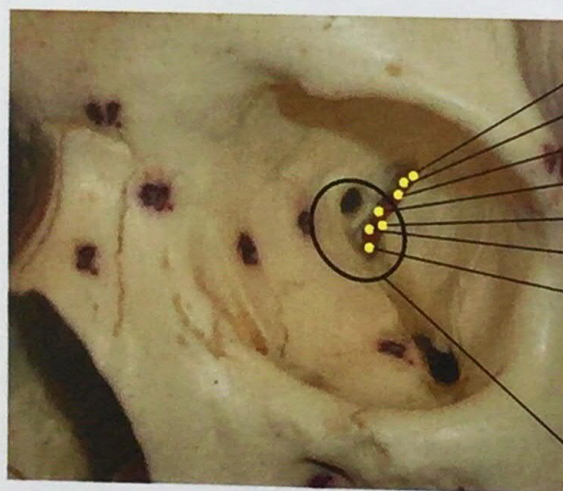


Questions: Please note that larger versions of these images are available in appendix 1

1. What structures pass through the superior orbital fissure and in what configuration?
2. What and where is Meckel's cave; what does it contain?
3. What is opening A? What passes through it?
4. What is opening B? What passes through it?
5. What is bone C?
6. Where is "Bill's bar" and how does it relate to the cranial nerves?

Answers:

1. What structures pass through the superior orbital fissure and in what configuration?



Lacrimal n. (from CN V ₁)	<i>Lateral Part</i>
Frontal n. (from CN V ₁)	
Trochlear n.	
Oculomotor n. (Sup. Div ⁿ)	<i>Medial Part</i>
Oculomotor n. (Inf. Div ⁿ)	
Nasociliary n. (from CN V ₁)	
Abducens n.	
<i>Other Contents:</i>	
-Meningeal branch of lacrimal a.	
-Branch of middle meningeal a.	
-Superior and inferior ophthalmic v.	
Annulus of Zinn*	

*Fibrous band (continuous with middle cranial fossa dura); origin of the four rectus muscles.

2. What and where is Meckel's cave; what does it contain?

A CSF filled pouch formed by two layers of dura which contains the trigeminal ganglion. It is located immediately posterolateral to the cavernous sinus. The cavernous carotid artery lies immediately medial to the trigeminal ganglion.

3. What is opening A? What passes through it?

Foramen ovale. Remember the mnemonic "OVALE": Otic ganglion, V₃, Accessory meningeal artery, Lesser petrosal nerve, Emissary veins).

4. What is opening B? What passes through it?

Hypoglossal canal. Hypoglossal nerve.

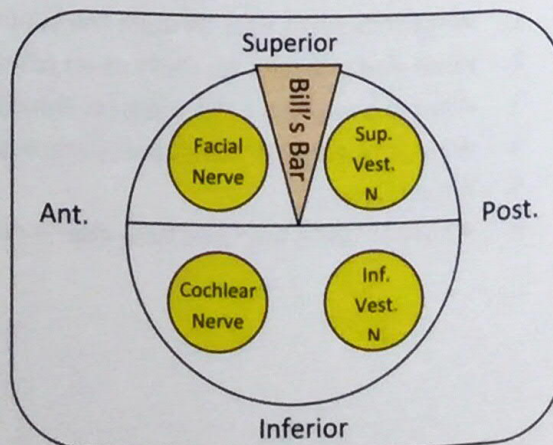
5. What is bone C?

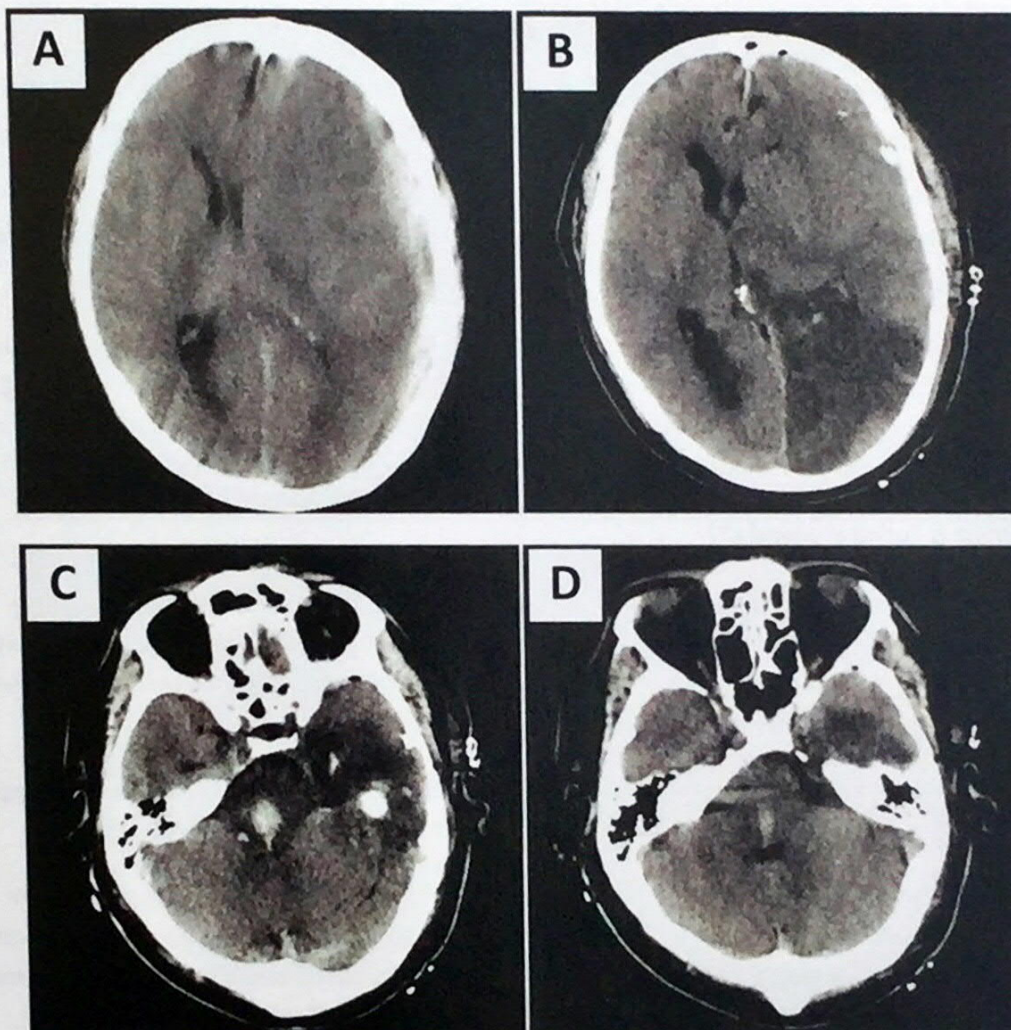
Sphenoid bone; greater wing.

6. Where is "Bill's bar" and how does it relate to the cranial nerves?

Bill's bar is a bony spur seen within the internal auditory canal, it divides the canal into anterior and posterior portions.

Remember "7up, Coke down"





This 47 year old is brought to A&E having been found unconscious in the street. On arrival in A&E his GCS is 4 ($E_1V_1M_2$). His pupils are 6mm and sluggishly reactive to light at best. He is intubated, ventilated and taken to CT for a scan (image A).

Questions:

1. What is the diagnosis? How would you manage this patient surgically?
2. Following surgery, ICP readings remain <20 mmHg. Temperature is 40°C . GCS on sedation hold is 3. How would you proceed?
3. What is "central pyrexia"?
4. After one week on ICU, there is no change clinically. A repeat scan is performed (images C-D). The pupils have remained fixed and dilated since surgery. What now?
5. What does brainstem testing involve?
6. Who may perform brainstem testing?

Answers:

- 1. What is the diagnosis? How would you manage this patient surgically?**

Left Acute subdural haematoma. This patient should undergo urgent craniotomy and evacuation of haematoma. If the brain is anything other than 'slack', craniectomy and ICP monitor insertion should be considered.

- 2. Following surgery, ICP readings remain <20 mmHg. Temperature is 40°C. GCS on sedation hold is 3. How would you proceed?**

Investigate extensively for causes of low GCS and high temperature, other than primary insult. Post operative imaging to exclude secondary insult/rebleed/hydrocephalus. Full septic screen. Consider cardiovascular and endocrinological dysfunction, including hyponatraemia. EEG should be performed to exclude seizure-related activity.

- 3. What is "central pyrexia"?**

Persistent, fever (>38°C) despite no other cause (sepsis) in the context of brain insult. Mechanism is debatable but is most commonly seen in hypothalamic compression. Central pyrexia is more commonly observed in haemorrhagic brain insult than in ischaemic events. There has been suggestion that the presence of bloodied CSF is a contributory factor.

- 4. After one week on ICU, there is no change clinically. A repeat scan is performed (images C-D). The pupils have remained fixed and dilated since surgery. What now?**

The repeat scan shows new posterior territory and temporal infarction, pons/midbrain haemorrhage and surrounding ischaemia. Assuming that other causes are excluded (see answer 2), it is reasonable to consider the withdrawal of care for this patient if brain stem death is confirmed. This decision should be made by the MDT, involving the family.

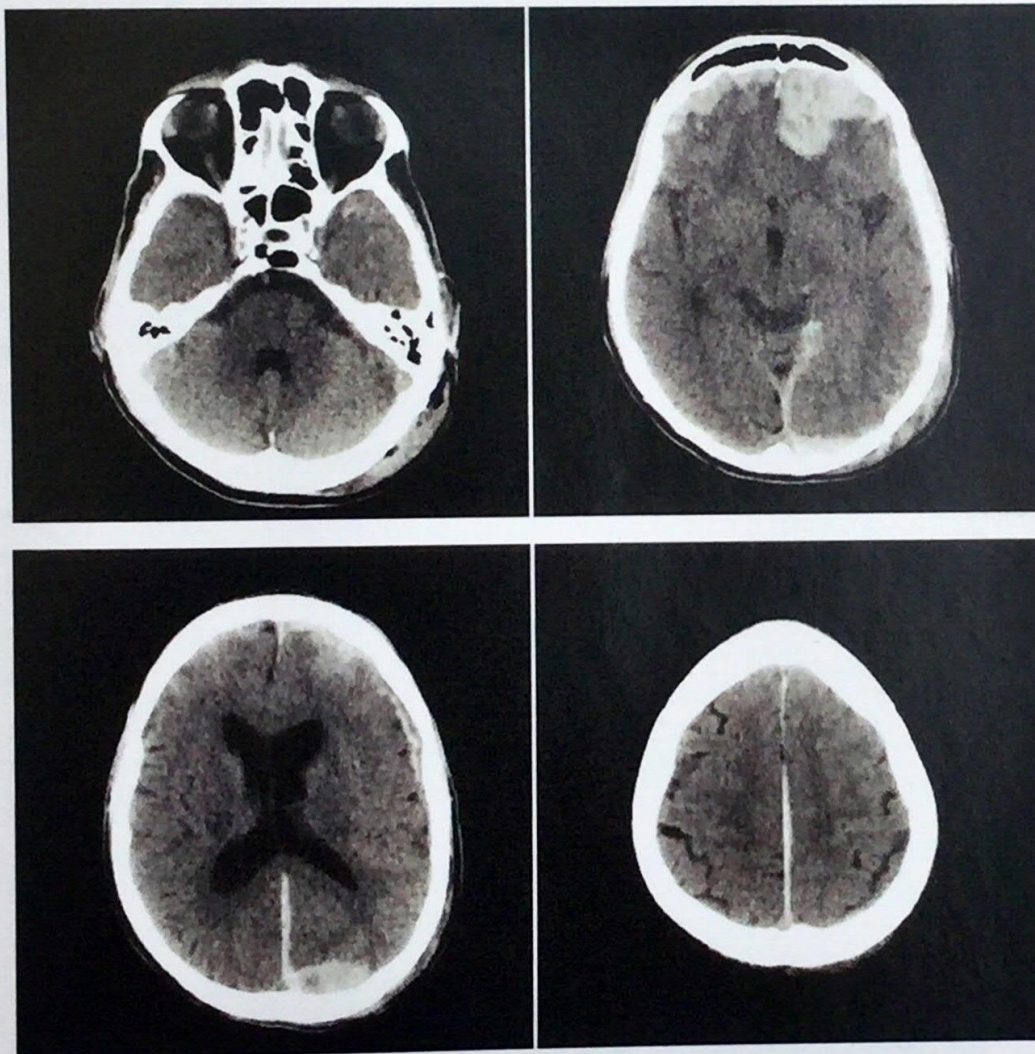
- 5. What does brain stem testing involve?**

Suitability for brain stem testing requires: consensus on severity of injury and that it is due to irreversible brain damage, absence of effects of depressant drugs, absence of hypothermia, all reversible causes of depressed GCS (endocrinological, metabolic and cardiovascular) excluded and all reversible causes of apnoea excluded

Brain stem death confirmed if: pupils fixed and dilated, absent corneal reflexes, absent oculovestibular reflexes, no response to supraorbital pressure, no gag reflex/response to bronchial stimulation with tracheal suction, no respiratory effort in response to disconnection from the ventilator for long enough for pCO₂ to reach ≥6.0 kPa

- 6. Who may perform brainstem testing?**

At least two doctors should each perform testing. Both must have been GMC registered for ≥5 years. At least one must be a consultant. The testing doctors must have experience in the field (e.g. anaesthetists, neurosurgeons) and must not be on the transplant team.



This 44 year old man is admitted following a high-speed road traffic collision. His GCS on arrival in A&E is 8. He is intubated and ventilated for CT scan.

Questions:

1. Does this patient need an ICP monitor? Are there any guidelines on who does?
2. What is the gold-standard method for ICP monitoring?
3. If used, where should a parenchymal ICP monitors be placed?
4. Draw and annotate a 'normal' ICP waveform
5. How does intracranial pressure vary with contusional volume? Explain the relationship
6. What medical treatments can be used to treat raised ICP?
7. What is the DECRA Trial?

Answers:

1. Does this patient need an ICP monitor? Are there any guidelines on who does?

Yes. Brain Trauma Foundation Guidelines; monitor ICP if GCS ≤ 8 AND either: abnormal admission CT head or ≥ 2 risk factors for $\uparrow$ ICP (age > 40 , SBP < 90 mmHg, decerebrate or decorticate posturing).

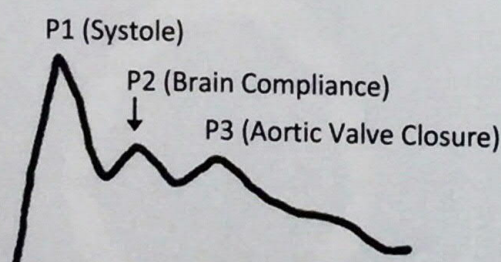
2. What is the gold-standard method for ICP monitoring?

Ventricular catheters are considered the gold-standard as they are the most accurate, allow for CSF drainage, and can be re-zeroed. They are more invasive than other methods and the risk of ventriculitis, difficulty of insertion and potential for blockage must be considered.

3. If used, where should a parenchymal ICP monitors be placed?

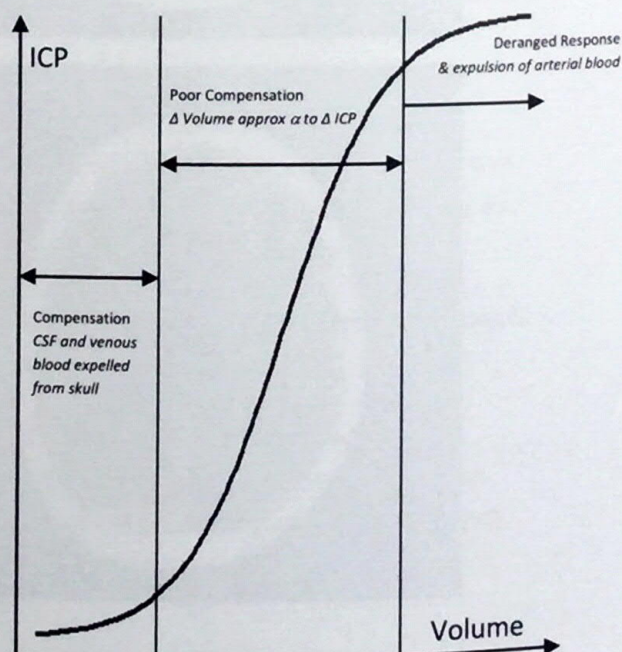
If diffuse injury; non-dominant frontal lobe.
If focal injury; peri-lesional parenchyma.

4. Draw and annotate a 'normal' ICP waveform:



**5. How does ICP vary with contusional volume?
Explain the relationship**

See right.



6. What medical treatments can be used to treat raised ICP?

Head up 30° . Exclude venous compression (e.g. from collar). Sedation. Hyperventilation ($p\text{CO}_2$ 4.5-5.0 kPa). Diuretics (e.g. Mannitol). Paralysis/ coma. *Note Mannitol's actions: osmotic diuresis, free radical scavenger, improves cell rheology, affects BBB permeability.*

7. What is the DECRA Trial?

International multicentre study. 155 pts randomised to bifrontotemporoparietal craniectomy or standard care if ICP failed to respond to first line therapy (> 15 mins with ICP > 20 mmHg). Craniectomy group results showed: Less time with high ICP, fewer interventions for ICP and fewer days on ICU but worse scores on the Extended GOS and greater risk of unfavorable outcome. 6 month mortality; craniectomy (19%) standard-care (18%). *Note: 27% bilaterally unreactive pupils% in craniectomy group, 12% in standard care group.*



Questions:

1. What is the embryological origin of the CNS?
2. What is neurulation? When does it occur?
3. At what level is the conus medullaris in the newborn and in the adult?
4. What is a neuroenteric cyst? How do they arise?
5. How is the pituitary gland formed?
6. What is an arachnoid cyst? How common are they?

Answers:

1. What is the embryological origin of the CNS?

The neural plate (derived from differentiated ectoderm) forms at week 3 and invaginates to form the neural tube and neural crest. The neural tube becomes the CNS and the neural crest becomes the PNS and ANS.

2. What is neurulation? When does it occur?

Neurulation occurs during week 4 of pregnancy and describes the formation of the neural tube.

During neurulation, the neural plate invaginates to form two neural crests which then fuse to form a neural tube with a central neural canal with openings at the rostral (anterior neuropore) and caudal (posterior neuropore) ends. The anterior neuropore closed at day 25 and the posterior neuropore at day 27. The neural tube becomes the brain (rostral $\frac{2}{3}$) and the central canal of the spinal cord (caudal $\frac{1}{3}$).

3. At what level is the conus medullaris in the newborn and in the adult?

This terminates at L3/4 in neonates, rising to L1/2 by adulthood.

4. What is a neuroenteric cyst? How do they arise?

Neuroenteric cysts are foregut duplication cysts that may connect to the spinal dura or even the spinal canal. The cysts arise from a failure of separation of the embryonic notochordal plate and presumptive endoderm during the third week of human development.

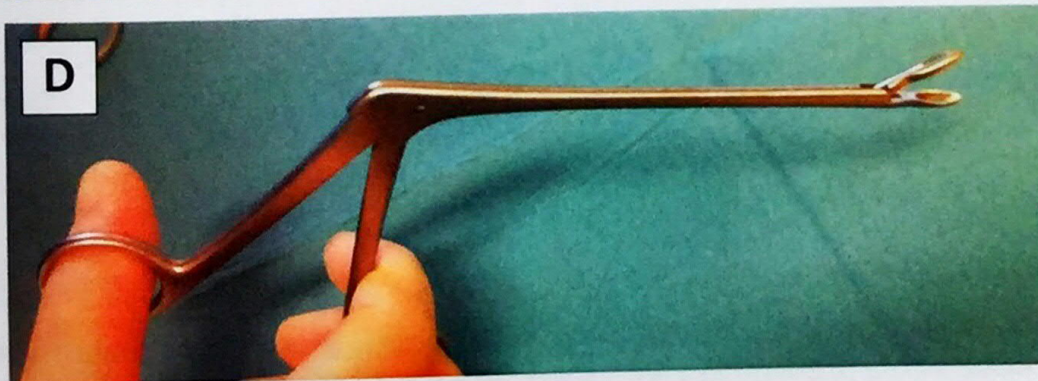
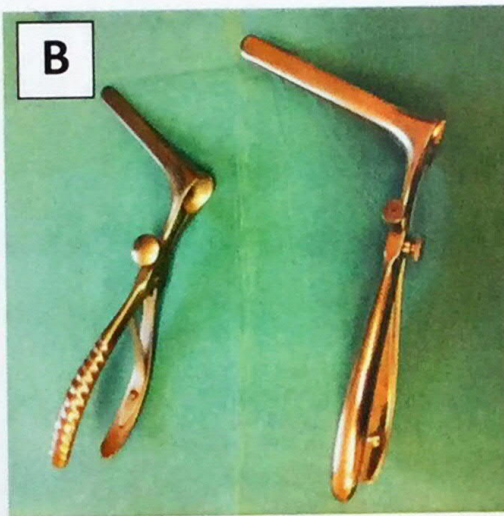
5. How is the pituitary gland formed?

The pituitary gland forms from two sources: an upgrowth of ectoderm known as the hypophysial pouch forms the anterior pituitary and a downgrowth of neuroectoderm from the diencephalon known as the neurohypophysial bud forms the posterior pituitary.

6. What is an arachnoid cyst? How common are they?

Arachnoid cysts are congenital, loculated collections of CSF that do not communicate with the ventricles. Middle cranial fossa lesions are the most common. Lesions are thought to arise from abnormal splitting of the arachnoid layer, although some theories suggest cortical hypoplasia may be the primary problem.

Post-mortem studies approximate the prevalence of arachnoid cysts to be in the region of 0.5%.



Questions:

1. Name the instruments shown above
2. How commonly do pituitary tumours secrete the various hormones?
3. What conditions/complications is acromegaly associated with?
4. What are the "Pituitary Stalk" and "Hook" effects?
5. What is Nelson's syndrome?
6. How is Cushing's syndrome investigated?

Answers:

1. Name the instruments shown above

A: Hardy (Cushing) bi-valve specula. **B:** Killian nasal specula. **C:** Pituitary instruments including: "Up/Down" and "Side-to-Side" knives, "Hockey stick" and curettes (various angulations). **D:** Tilley-Henckel ethmoid punch.

2. How commonly do pituitary tumours secrete the various hormones?

PRL (40%), non-secreting (20%), GH (20%), ACTH (15%), others & multiple (15%).

3. What conditions/complications is acromegaly associated with?

Arthropathy, cardiac disease (includes heart failure), diabetes, malignancy (esp. colonic), neuropathy, sleep apnoea (poor pharyngeal muscle tone). *Some reversibility with ↓GH.*

4. What are the "Pituitary Stalk" and "Hook" effects?

Pituitary Stalk Effect: Dopamine has an inhibitory feedback effect on prolactin production. Blockage of the dopamine delivery system e.g. tumour or trauma results in hyperprolactinaemia. Serum levels of >200 ng/ml are almost certainly due to prolactinoma, rather than stalk effect.

Hook Effect: If prolactin levels are very high, this may impair the antigen-antibody binding process in an immunoassay, potentially resulting in a false -ve.

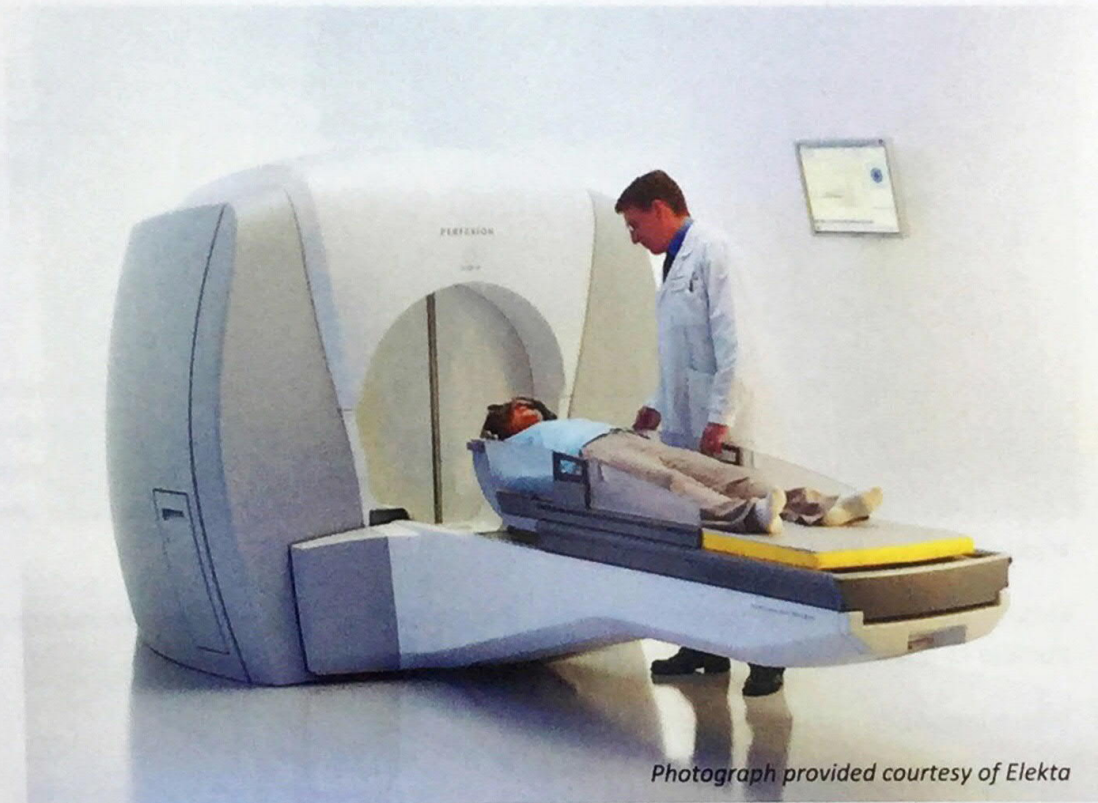
5. What is Nelson's syndrome?

The absence of cortisol following bilateral adrenalectomy causes loss of ACTH negative feedback and uninhibited growth of a pre-existing pituitary adenoma. ACTH is derived from proopiomelanocortin which is also a precursor of melanocyte stimulating hormone (MSH) and therefore levels of MSH also rise. Patients may experience visual problems (due to pituitary mass effect), muscle weakness and skin hyperpigmentation (due to excess MSH).

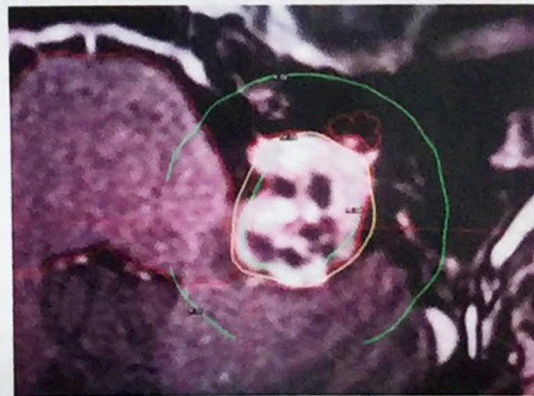
6. How is Cushing's syndrome investigated?

The commonest cause of Cushing's syndrome is iatrogenic (drugs). Excluding this, patients should initially be investigated with 24-hour cortisol measurements or low-dose dexamethasone suppression test. If these prove positive, late afternoon serum ACTH measurements can be used to determine whether the source is likely pituitary or otherwise. If a pituitary source is indicated, patients should undergo MRI scan to assess further. If not, then patients should be screened (CT chest/abdomen) for other sources. Inferior petrosal sinus sampling may be employed in cases of microadenoma or of inconclusive MRI results.

For an example algorithm flow-chart, see appendix 2



Photograph provided courtesy of Elekta



This patient is undergoing Gamma Knife treatment for a vestibular Schwannoma.

Questions:

1. What is Gamma Knife®? How does it work?
2. Which lesions are suitable for Gamma Knife?
3. Which lesions do not respond to Gamma Knife?
4. What is pseudoprogression?
5. What are the absolute contraindications to stereotactic radiosurgery?

Answers:

1. What is Gamma Knife®? How does it work?

Gamma Knife® is a type of stereotactic radiosurgery in which patients receive gamma radiation from 201 individual cobalt-60 isotopes, arranged in a spherical array, in which the patient's head is centred. The patient's head is secured in a frame and the tumour margins defined such that the beams of radiation converge in the target area, sparing the surrounding normal brain which only receives very low doses of gamma radiation.

2. Which lesions are suitable for Gamma Knife?

Can be used for a wide range of cranial pathology, most commonly: vestibular Schwannoma, meningioma, cerebral metastasis and glioma. Has also been used in many others, including tumours of the pituitary, choroid plexus papilloma, ependymoma, medulloblastoma, pineal tumours etc. There is also a role in treating vascular anomalies, including AVM and for use in trigeminal neuralgia.

Lesions should be <3cm in maximal diameter. There is usually also a policy on the maximum number of lesions that can be treated, in Leeds patients must have ≤3 lesions.

3. Which lesions do not respond to Gamma Knife?

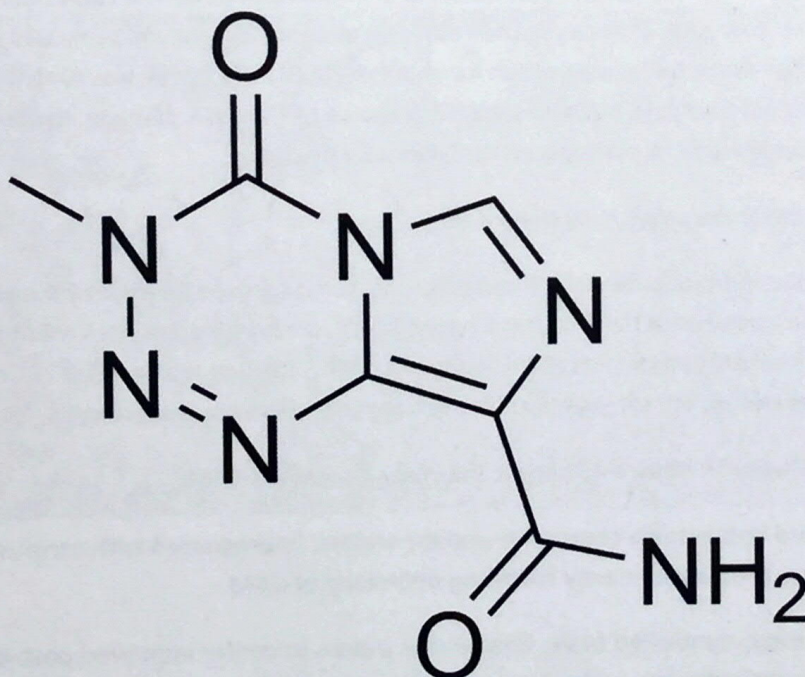
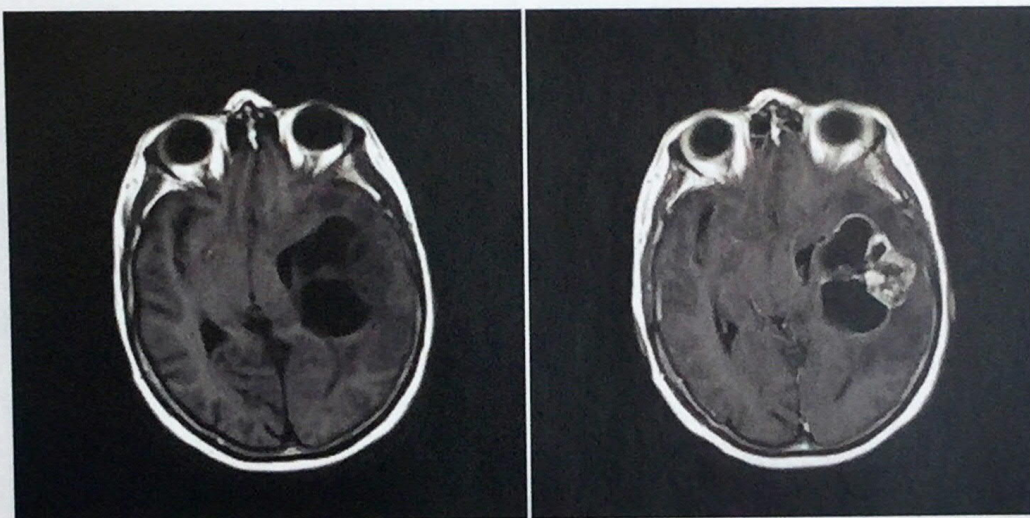
Gamma Knife cannot be used for spinal tumours. Besides lesions that are too large or too numerous; there are relatively few brain tumours that are not suitable for Gamma Knife treatment. It is not advisable to treat a single presumed metastasis with Gamma Knife if the primary tumour is not known. There are some series to suggest that Gamma Knife can improve survival in selected cases of glioblastoma multiforme, although it is not a curative treatment option. The role of Gamma Knife for cavernoma remains debatable.

4. What is pseudoprogression?

This is the phenomenon whereby apparently progressive and enhancing lesions are identified on MRI immediately following the end of treatment (such as temozolomide therapy), which are not related to true tumour progression, rather local necrosis and oedema that are caused by the treatment itself.

5. What are the absolute contraindications to stereotactic radiosurgery?

Patients should satisfy the usual MRI safety requirements (no pacemaker etc). Patients who will not be able to tolerate the surgery (remain still for long enough, tolerate fitting of the frame etc.) cannot undergo surgery. Claustrophobic patients may require conscious sedation. Young, female patients undergo pregnancy testing prior to Gamma Knife treatment, although any dose that an unborn foetus might receive should be negligible, in theory.



This gentleman's histological diagnosis is confirmed as glioblastoma multiforme (WHO grade IV).

Questions:

1. What is the Stupp protocol? What is its evidence?
2. By what mechanism does fractionated radiotherapy work?
3. What is temozolomide? How does it work?
4. What is Gliadel®? What evidence is there for its use in GBM?
5. What is meant by the term "Performance Status"?

Answers:

1. What is the Stupp protocol? What is its evidence?

During the Stupp study (Stupp et al. *NEJM* 2005; **352**: 987-996) 573 patients were randomised to radiotherapy (60 Gy total: 2 Gy x 5 days/week x 6 weeks) ± concurrent daily temozolomide treatment (75mg/m² body surface area) throughout the 6 weeks of radiotherapy, followed by 6 further cycles of temozolomide (150-200mg/m² body surface area x5 days in each 28-day cycle).

This paper showed that the **early** addition of temozolomide to radiotherapy statistically improves median survival (12.1 months to 14.6 months, $p < 0.001$) as well as median progression-free survival (5 months to 6.9 months, $p < 0.001$).

2. By what mechanism does fractionated radiotherapy work?

High energy x-rays are accelerated towards the radiation field. The radiation is delivered in lower dose 'fractions' that cause DNA damage to cells. Tumour cells are less able to repair any initially non-lethal damage than normal cells and are thereby less likely to survive. The unrepaired tumour cells become hypoxic because of the DNA damage, making them yet more radiosensitive to subsequent radiotherapy fractions.

3. What is temozolomide? How does it work?

Temozolomide (molecular structure shown on question page) is an oral chemotherapeutic that depletes stores of the DNA repair enzyme O⁶-methylguanine-DNA methyltransferase (MGMT). Temozolomide acts as an alkylating agent; causing methylating MGMT promoters, thereby depleting MGMT stores and inhibiting DNA repair in tumour cells.

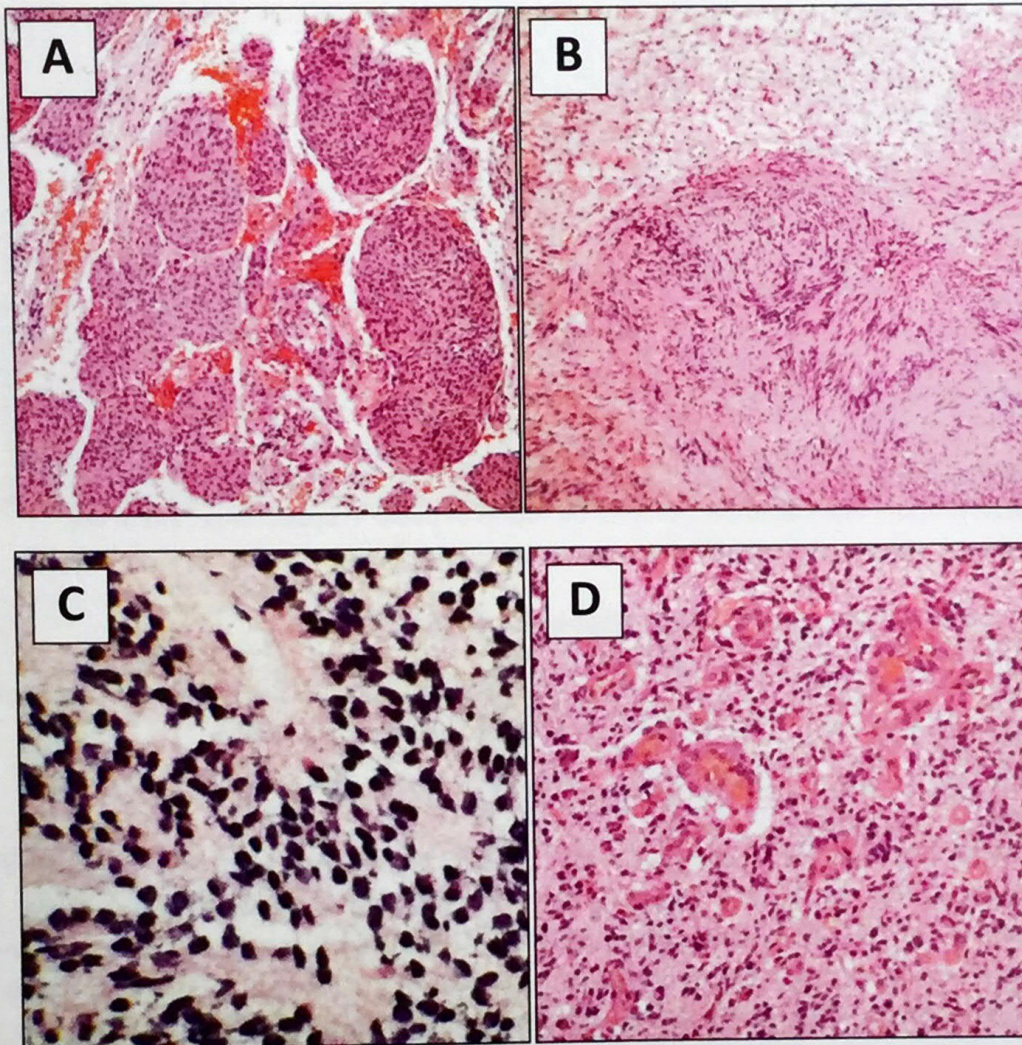
4. What is Gliadel®? What evidence is there for its use in GBM?

Gliadel® are implantable chemotherapeutic wafers, impregnated with carmustine. These are used to line the tumour cavity following debulking of GBM.

In two placebo-controlled trials, Gliadel was shown to confer improved post-operative survival in patients undergoing initial debulking for GBM. A similar trial assessed survival in patients undergoing surgery for tumour recurrence and showed no benefit is associated with Gliadel insertion.

5. What is meant by the term "Performance Status"?

Performance status is a way to quantify a patient's general health. It is used clinically to assess a patient's suitability for oncological treatment and in research to approximate quality of life outcome. One example of a performance status scoring system is that of Karnofsky, which is a scale from 100% (normal function) to 0% (dead) ⇒ Karnofsky DA et al. *The clinical evaluation of chemotherapeutic agents in cancer*. In: MacLeod CM (ed.): *Evaluation of chemotherapeutic agents*. New York: Columbia University Press 1949; 191-205



Questions:

1. Describe the features of slide A. What is the diagnosis?
2. Describe the features of slide B. What is the diagnosis?
3. Describe the features of slide C. What is the diagnosis?
4. Describe the features of slide D. What is the diagnosis?
5. How is a "frozen section" performed?

Answers:

1. Describe the features of slide A. What is the diagnosis?

Slide A shows the whorling and lobules of uniform cells that are characteristic of meningiomas.

2. Describe the features of slide B. What is the diagnosis?

Slide B shows elongated spindle cells arranged in regions of densely cellular (Antoni A) tissue or looser, hypocellular tissue (Antoni B). These features are characteristic of a (vestibular) schwannoma.

3. Describe the features of slide C. What is the diagnosis?

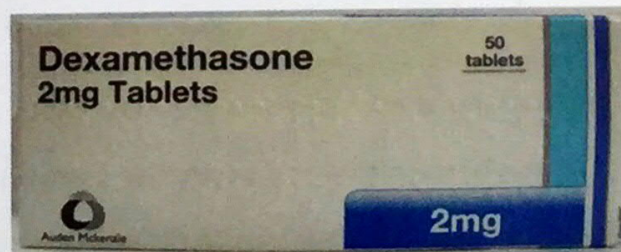
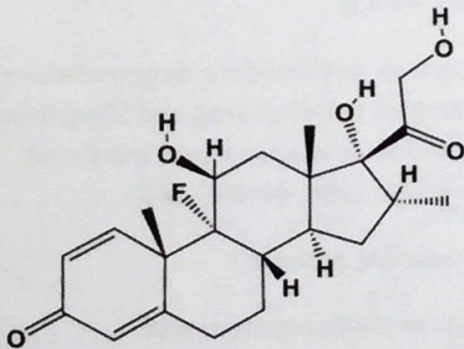
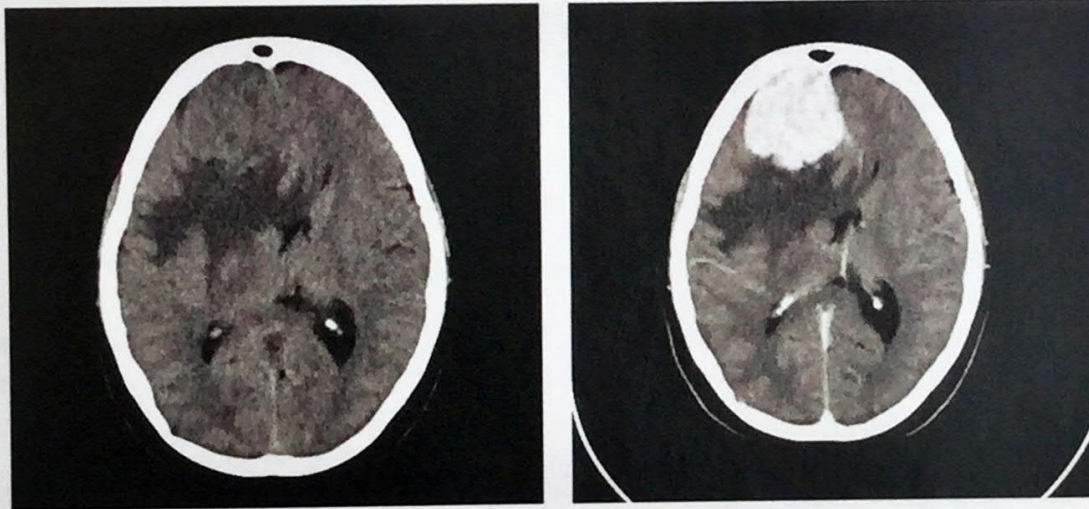
Slide C shows hypercellular tissue with benign-looking cells and little evidence of proliferation. Cell nuclei are noted to be round or ovoid in appearance. There are pseudorosettes present, which are typical of ependymoma.

4. Describe the features of slide D. What is the diagnosis?

Slide D shows extremely vascular tissue with high proliferation rates. There are areas of necrosis and evidence of vascular proliferation. The diagnosis is GBM.

5. How is a "frozen section" performed?

A frozen section is a technique that is employed to give a rapid histological diagnosis of a specimen. It is less accurate than formalin-fixed analysis. Tissue is embedded in a poly ethylene glycol and polyvinyl alcohol gel. A *cryostat* is used to rapidly freeze the specimen to between -20 and -30°C before it is sliced in sections of $5-10 \times 10^{-6}$ m thickness. The slices are stained (usually H&E) and mounted on slides. The whole technique can take as little as 10 minutes to perform.



Product image kindly provided by:
Auden Mckenzie (pharma Division)

This 60 year old lady is admitted with personality changes and drowsiness. Pre-operatively she is commenced on dexamethasone "To reduce the swelling in the head".

Questions:

1. What dosage, frequency and duration of treatment should the patient receive?
2. What is the evidence for steroids in cerebral oedema secondary to tumours?
3. What is dexamethasone's mechanism of action in reducing oedema?
4. What is the method of absorption, distribution, metabolism, excretion and half life of oral dexamethasone?
5. What are the potential complications of dexamethasone use in this patient?

Answers:

1. What dosage, frequency and duration of treatment should the patient receive?

Steroid regimens vary between units. One typical starting dose would be 16mg of oral dexamethasone once daily (in the morning) with a gradual reducing dose post-operatively. Some units favour intravenous usage. During the course of steroids the patient would typically receive gastric protection (e.g. oral ranitidine, 150mg twice daily).

2. What is the evidence for steroids in cerebral oedema secondary to tumours?

There is no RCT evidence for dexamethasone usage versus placebo in this patient group; however it is a well-established treatment for cerebral oedema. The principle for the use of steroids is a crossover from animal studies.

An observational study of 14 patients documented favourable outcomes in patients treated with dexamethasone (Galicich et al. *J Lancet* 1961; **81**: 46-53).

A double blind randomised controlled trial comparing high and low dose dexamethasone in metastatic brain tumours found no difference in outcomes between 4mg and 16mg doses (in patients with normal GCS and no evidence of raised intracranial pressure) but found greater side effects in higher doses (Vecht et al. *Neurology* 1994; **44**: 675-680).

3. What is dexamethasone's mechanism of action in reducing oedema?

Dexamethasone is a synthetic glucocorticoid and is an anti-inflammatory and immunosuppressant drug. It is thought to reduce brain tumour-associated vascular permeability thereby reducing cerebral oedema.

4. What is the method of absorption, distribution, metabolism, excretion and half life of oral dexamethasone?

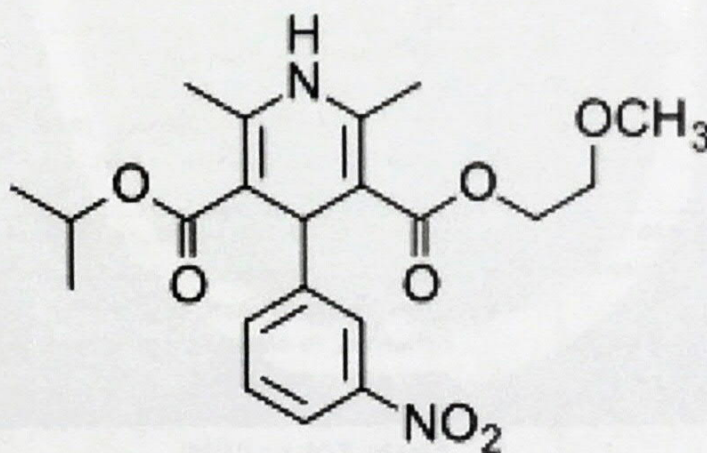
Absorption: absorption fraction 100% (stomach, small intestine, large intestine)
Distribution: blood-bound, maximum brain concentration at 3-4 hours post-administration
Metabolism: via hepatic microsomes
Excretion: renal excretion
Half Life: 56 hours

5. What are the potential complications of dexamethasone use in this patient?

Side effects include: Cushingoid features, mood disorders, steroid-induced psychosis, GI upset, weight gain, impaired glucose tolerance, hypertension and sleep disturbance.

Middle to long term usage can lead to immunosuppression, osteoporosis, myopathy and skin breakdown.

It is worthy of note that dexamethasone may affect phenytoin or carbamazepine levels and these should be monitored during treatment.



A 38 year old man is admitted acutely following an aneurismal subarachnoid haemorrhage. He is started on regular nimodipine by the on-call neurosurgical SHO.

Questions:

1. What dosage, frequency and duration of treatment should the patient receive?
2. What is the evidence that this regimen will benefit the patient?
3. What is nimodipine's mechanism of action in vasospasm prevention?
4. Are there any contra-indications to nimodipine therapy?

Answers:**1. What initial dosage, frequency and duration of treatment should the patient receive?**

60mg of oral nimodipine every four hours for 21 days.

2. What is the evidence that this regimen will benefit the patient?

There is class I evidence for nimodipine prophylaxis in the prevention of cerebral vasospasm for patients who have experienced aneurismal subarachnoid haemorrhage. The British aneurysm nimodipine trial (Pickard et al. *BMJ* 1989; **298**: 636-642) is a prospective, double-blinded, multi-centre randomised controlled trial of nimodipine versus placebo:

Parameter	Further Details
Subjects	• 554 (278 nimodipine, 276 placebo)
Inclusion Criteria	• <96 hours since ictus • Proven SAH (CT or LP positive) • Age > 18 years
Exclusion Criteria	• ≥96 hours since ictus • Pregnancy • Major co-morbidities (hepatic, renal, lung, cardiac) • Coma due to SAH within 7 days • Lack of consent
Primary Endpoint	• Rate of cerebral infarction, re-bleed or poor outcome
Secondary Endpoints	• Glasgow Outcome Score ≥ 3 months after ictus • Other causes of death and disability including: initial ischaemia, re-bleeding, haematoma, hydrocephalus, operative complications
Results	• Reduced incidence of vasospasm (nimodipine 22%, placebo 33%, $p = 0.014$) • Reduced incidence of poor outcome (nimodipine 20%, placebo 33%, $p < 0.001$)

There is further evidence from meta-analysis of 6 trials and a Cochrane review from 2007.

3. What is nimodipine's mechanism of action in vasospasm prevention?

Nimodipine is a calcium channel blocker. It impairs the influx of calcium ions into cardiovascular smooth muscle cells. The effects of nimodipine are more pronounced in cerebrovascular tissue than other vessels. It has been suggested that this is due to the fact that the molecule is lipophilic, allowing it to cross the blood-brain barrier relatively freely.

4. Are there any contra-indications to nimodipine therapy?

Contra-indications to nimodipine include: use within 1 month of myocardial infarction, unstable angina and acute porphyria. Use in pregnancy is advised if potential benefits outweigh risks. There is no available evidence of effects when breast feeding. Interestingly, it should not be used concurrently with phenobarbital, phenytoin or carbamazepine.



You are asked to review the imaging for this patient who has been admitted via A&E with seizures.

Questions:

1. Describe the contrast scan shown. What are the differential diagnoses for this lesion?
2. What are the radiological differences between high grade glioma and abscess on CT?
3. What further imaging would you request to exclude abscess?
4. How does this technology work?
5. What is Pott's puffy tumour?

Answers:

1. Describe the imaging shown. What are the differential diagnoses for this lesion?

This axial CT image of the head shows a large, left-sided lesion with presumed peripheral enhancement (no pre-contrast image is provided for comparison). The lesion lies with the left frontal lobe and is associated with a substantial amount of oedema. There is some associated distortion of the lateral ventricles.

The two main differential diagnoses for this lesion are abscess and tumour (high grade intrinsic or metastatic).

2. What are the radiological differences between high grade glioma and abscess on CT?

If the inner wall of the enhancing rim of the lesion is smooth and regular in appearance, the lesion is more likely to be abscess than tumour. Abscesses may also have small, adjacent satellite lesions around their periphery. There may also be evidence of infection elsewhere, such as ventriculitis.

3. What further imaging would you request to exclude abscess?

This patient should undergo urgent MRI assessment to further assess this lesion. In particular, a diffusion-weighted sequence should be included.

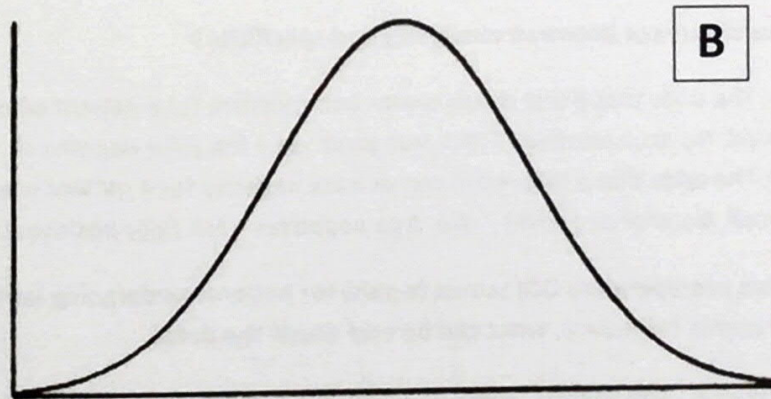
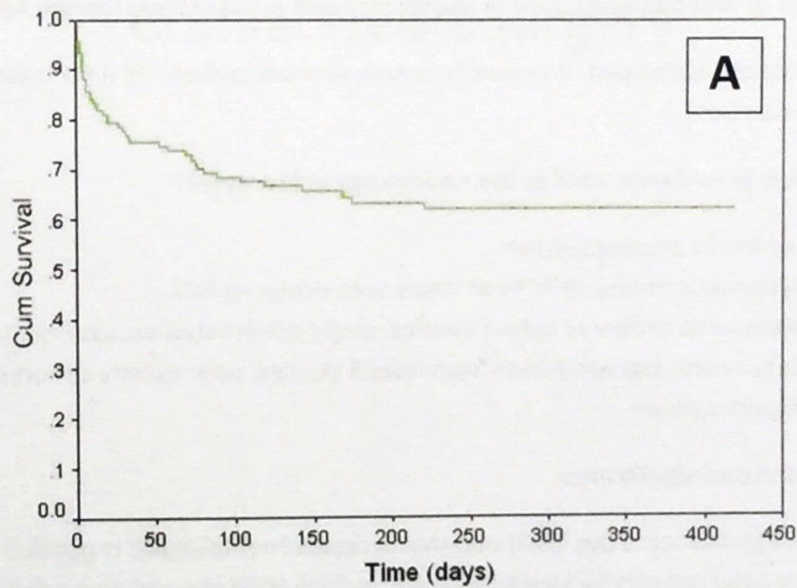
In abscess, high DWI signal is usually present centrally, implying restricted diffusion. Conversely, there is typically low signal on ADC mapping.

4. How does this technology work?

DWI allows for comparison of the relative amounts of Brownian motion of water molecules within tissues. Within the thin, cystic fluid of a GBM, water molecules are relatively free to move, however the thick pus within an abscess restricts movement substantially.

5. What is Pott's puffy tumour?

This is the presence of a visible forehead swelling due to frontal sinusitis that has caused osteomyelitis and eroded into the subgaleal space. Concurrent retrograde erosion may result in the presence of a simultaneous brain abscess.



Questions:

1. What type of plot has been used in the shunt audit presentation (image A)?
2. What levels of evidence exist in the neurosurgical literature?
3. Define statistical significance.
4. What is the difference between sensitivity and specificity?
5. Plot B shows pre-operative ODI (Oswestry disability index) scores (x-axis) for patients undergoing laminectomy. If the y-axis represents frequency, what can be said about the data?
6. What are type I and type II errors?
7. What test would be suitable to compare the pre and post-operative ODI scores for patients undergoing laminectomy?

Answers:

- 1. What type of plot has been used in the shunt audit presentation (image A)?**

This is a Kaplan–Meier plot. It is used to assess survival probability (in this case that of a shunt) versus time.

- 2. What levels of evidence exist in the neurosurgical literature?**

The NHS evidence grading system:

Level A: Systematic review of RCTs or single well-designed RCT

Level B: Systematic review of cohort studies, single cohort studies, case-control studies

Level C: Case series, extrapolations from level B studies, poor quality cohort studies

Level D: Expert opinion

- 3. Define statistical significance.**

Statistical significance is the likelihood that an experimental result is not due to chance. Commonly used cut-offs for significance are: $p < 0.05$ (<5% chance) and $p < 0.01$ (<1% chance).

- 4. What is the difference between sensitivity and specificity?**

Sensitivity: The odds that a test result comes back positive for a patient who has a condition.

- *Is calculated:* $\text{No. true positives} / (\text{No. true positives} + \text{No. false negatives})$.

Specificity: The odds that a test result comes back negative for a patient who is well.

- *Is calculated:* $\text{No. true negatives} / (\text{No. true negatives} + \text{No. false positives})$.

- 5. Plot B shows pre-operative ODI scores (x-axis) for patients undergoing laminectomy. If the y-axis represents frequency, what can be said about the data?**

This is the typical “bell-shaped” curve of a normally distributed (Gaussian) dataset. Some statistical tests are only valid for normally distributed variables.

- 6. What are type I and type II errors?**

Type I Error: The null hypothesis is rejected but it is, in fact, true. The consequence of a type I error is that something appears to have been proved that is actually not significant.

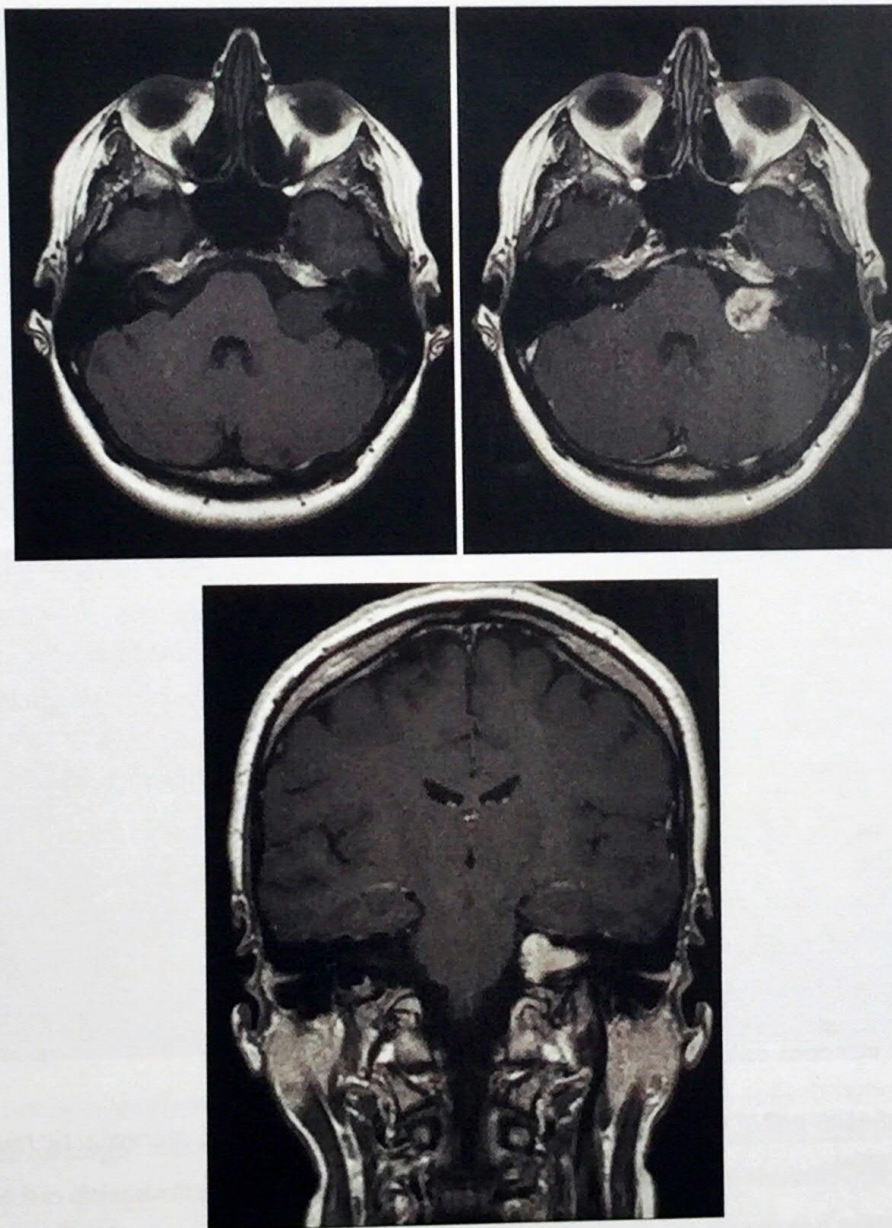
Type II Error: The null hypothesis is accepted but is, in fact, false. The consequence of a type II error is that something that is a significant finding is wrongly dismissed.

- 7. What test would be suitable to compare the pre and post-operative ODI scores for patients undergoing laminectomy?**

In this scenario, there are ‘paired’ datasets. The most appropriate test for this comparison would be the paired T-test.

Section 2:

Cranial Neurosurgery



This 51 year old was previously seen in clinic complaining of decreased hearing on left, tinnitus and unsteadiness. This one year follow-up scan shows increase in the size of the lesion; it is now 2cm in diameter. She complains of worsening headaches.

Questions:

1. Describe the images shown, what is the most likely diagnosis?
2. From where do these tumours arise?
3. Which presenting signs and symptoms are the most common?
4. What other tests might be requested prior to treatment?
5. What treatment options are available?

Answers:

1. Describe the images shown, what is the most likely diagnosis?

These are T1-weighted axial (top row) and coronal MRI images. The images show a discrete, contrast-enhancing lesion (top right) at the cerebello-pontine angle which has an apparently cystic central component. There appears to be an intracanalicular component. These features are typical of vestibular schwannoma (or acoustic neuroma).

2. From where do these tumours arise?

Most tumours arise from the Obersteiner-Redlich zone, approximately 1cm from where the nerve exits the brainstem. The most commonly affected nerve is the superior division of the vestibular nerve; hence the term vestibular schwannoma is favoured over acoustic neuroma.

3. Which presenting signs and symptoms are the most common?

Progressive unilateral (unless lesions are bilateral) hearing loss is the cardinal feature and is present in at least 95% of cases. Most patients with vestibular schwannoma complain of tinnitus, vertigo or both. Headache, otalgia and facial numbness and weakness are also recognised features but are far less commonly experienced. On examination, a patient would be Rinne's test positive (i.e. air conduction would be superior to bony conduction) and the patient would localise sound to the unaffected side in Weber's test.

4. What other tests might be requested prior to treatment?

A pure tone audiogram may be performed in order to document and monitor hearing loss although this is not essential. High-frequency hearing loss is the most common result for patients with a vestibular schwannoma.

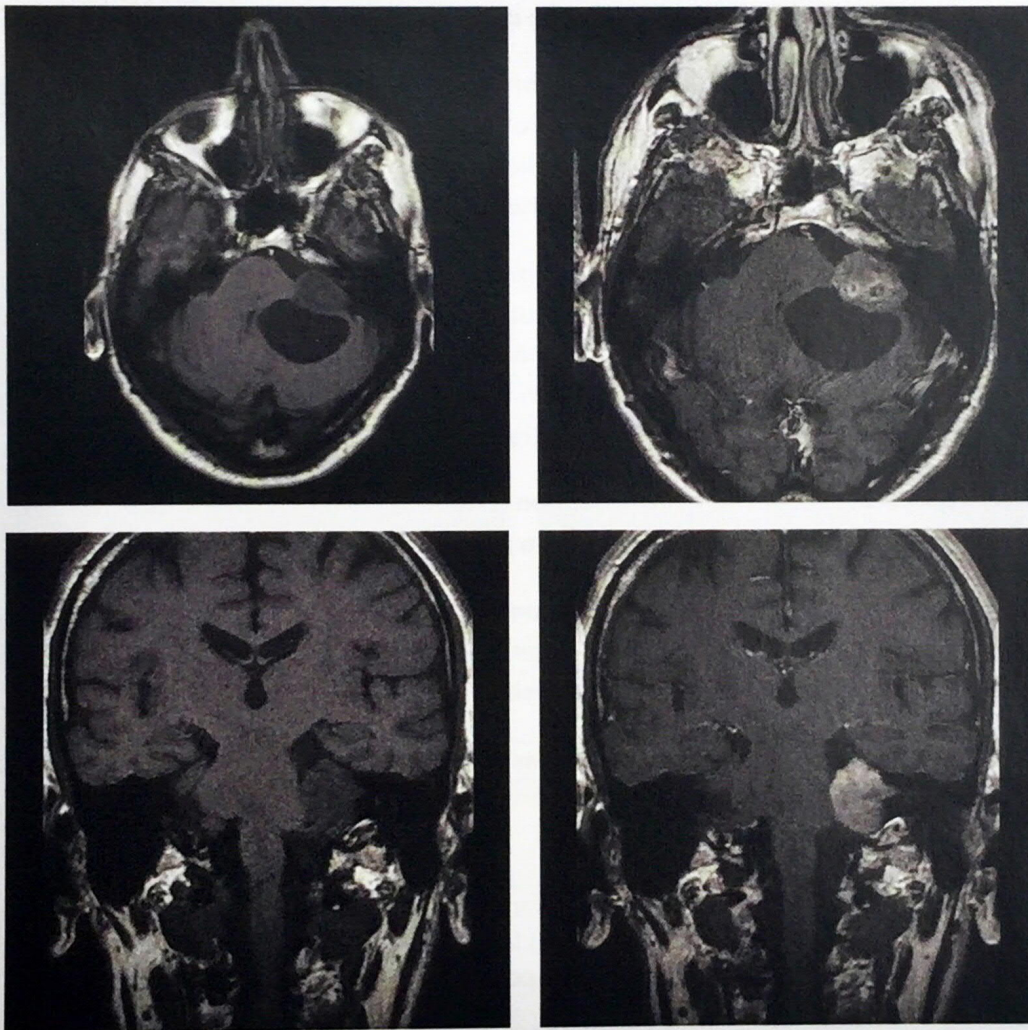
5. What treatment options are available?

The management of these lesions can be divided into: expectant, stereotactic radiosurgery and surgical.

This patient has shown relatively rapid tumour growth (most lesions grow at a rate of <10mm/year) and some symptom progression. Expectant management is probably inappropriate unless the patient refuses treatment.

Stereotactic radiosurgery may be considered in lesions with a maximal diameter of <3cm. Hearing preservation has been quoted at 50% in some series and the incidence of post-procedural complications (such as facial numbness, weakness and tumour swelling) is low. Long-term follow-up data (and therefore recurrence rates) are unknown.

Surgery offers the potential for complete tumour resection in up to 97% of cases and if this can be achieved recurrence rates may be as low as 5% over a 10 year follow-up period.



This 76 year old man is referred with a 12-month history of unsteadiness, speech slurring and dysarthria. He no longer has any useful hearing on the left side. He has become increasingly unsteady and has deteriorated following cessation of a recent trial of oral steroids. He has no prior medical history of note.

Questions:

1. Describe the images shown; what are the differentials for a tumour in this region?
2. The patient is very keen to have this lesion treated aggressively. Briefly describe the possible surgical approaches to this region and their relative merits.
3. Post-operatively the patient complains of facial weakness. How is this graded?

Answers:**1. Describe the images shown; what are the differentials for a tumour in this region?**

These are T1-weighted axial (top) and coronal (bottom) MRI images; shown with (right) and without (left) contrast. A part cystic, part solid lesion is seen in the left cerebello-pontine angle. There is contrast enhancement of the solid component. There is evidence of mass effect of the brainstem and cerebellum. There is no hydrocephalus.

Differentials include: vestibular schwannoma (note extension into the internal auditory meatus), cystic meningioma (although no obvious dural tail is seen), arachnoid cyst (unlikely given the solid component), epidermoid (consider in lesions with a solid and cystic component), dermoid (would expect lesion to be hyperintense on T1W imaging), metastasis, AVM (no vascular flow voids, atypical contrast enhancement pattern) or abscess.

2. The patient is very keen to have this lesion treated aggressively. Briefly describe the possible surgical approaches to this region and their relative merits.

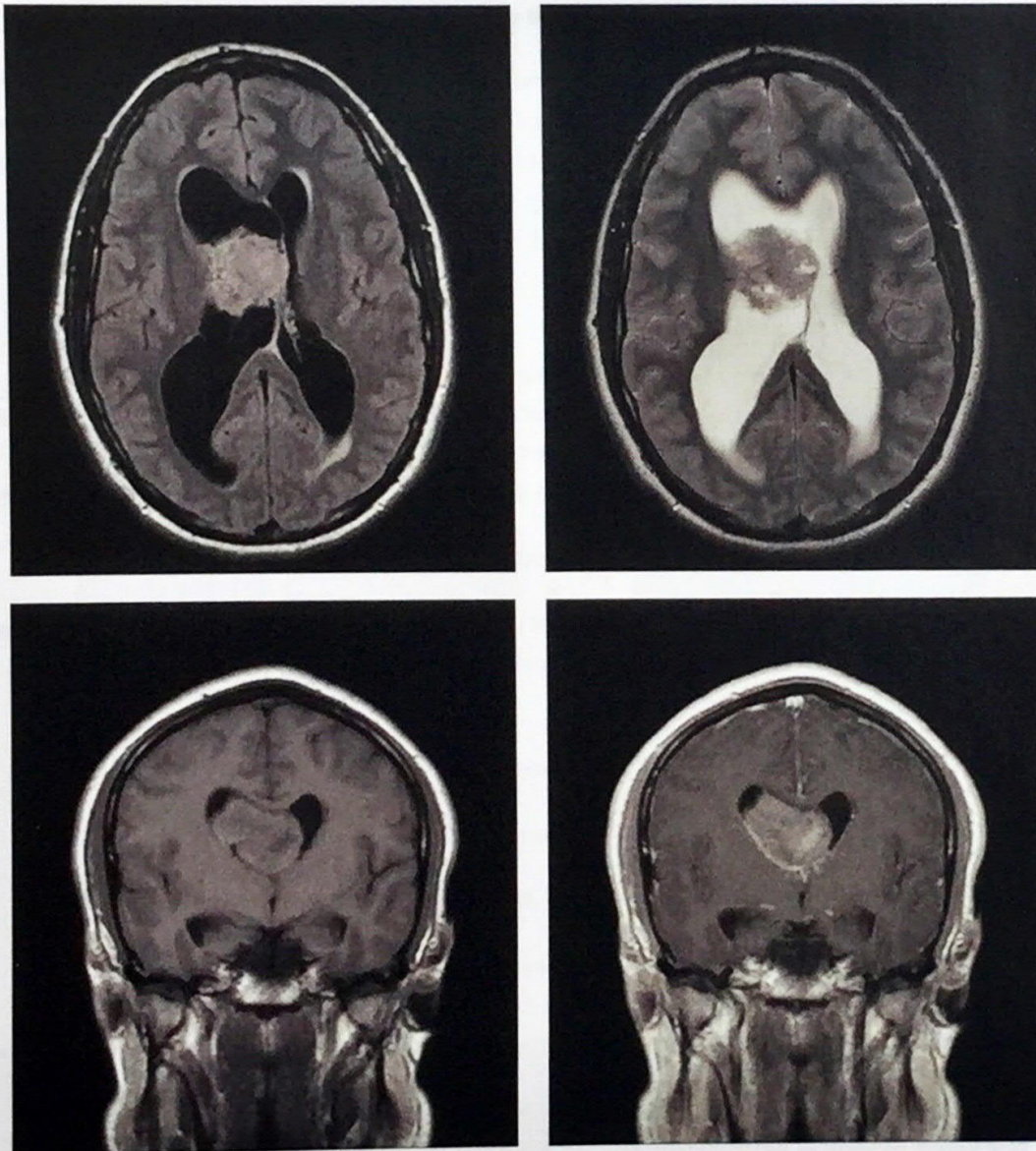
The three classical approaches to this region are listed below. Be aware that it is possible to combine the translabrynthine and retrosigmoid approaches in some cases. Also consider the role of stereotactic radiosurgery for any residuum seen post-operatively.

Approach	Advantages	Disadvantages
Retrosigmoid (Suboccipital)	Highest chance of hearing preservation	Late visualisation of facial nerve Higher post-operative morbidity
Translabrynthine	Early visualisation of facial nerve Useful for lesions with large intracannalicular component	Sacrifices any remaining hearing Difficult to excise larger lesions Increased surgical duration Poor access to CP angle
Middle Fossa (Extradural Subtemporal)	Potential for hearing preservation in intracannalicular lesions	Only suitable for small, laterally-located lesions Increased incidence of facial nerve damage

3. Post-operatively the patient complains of facial weakness. How is this graded?

Facial nerve dysfunction can be classified using the House-Brackmann scale (first described in 1985). It is graded as follows:

- Normal Function
- Mild Dysfunction (normal appearance at rest; subtle asymmetry on movement)
- Moderate Dysfunction (non-disfiguring asymmetry at rest but clear asymmetry on movement)
- Moderate-Severe Dysfunction (obvious asymmetry at rest and on movement)
- Severe Dysfunction (gross asymmetry; movements barely visible)
- Complete Paralysis



This 24 year old lady presents with headache and visual disturbance. On examination, she is neurologically intact but has papilloedema.

Questions:

1. Describe the images shown. What are the differential diagnoses?
2. How should this patient be managed?
3. What surgical approach would you advocate for this lesion?
4. What are the typical microscopic characteristics of central neurocytoma?
5. What immunohistochemical markers are displayed in central neurocytoma?
6. What is the MIB-1 index? What is the relevance in central neurocytoma?

Answers:

1. Describe the images shown. What are the differential diagnoses?

Axial T2 flair and T2 weighted (top row) and pre and post-contrast T1 coronal sequences (bottom row) showing a large, well-defined diffusely enhancing mass is demonstrated within the body of the right lateral ventricle complicated by hydrocephalus. This tumour is likely to be of ependymal origin, differentials being: central neurocytoma (as in this case), subependymoma, subependymal giant cell tumour or oligodendroglioma.

2. How should this patient be managed?

Given the history of papilloedema and visual deterioration, this patient needs urgent intervention, either to deal with the hydrocephalus alone or to attempt primary resection. Complete surgical resection offers the chance of cure. If this is not achieved/possible, adjuvant radiotherapy (either whole-brain or γ -knife) should be performed for residuum.

3. What surgical approach would you advocate for this lesion?

Most tumours arise from within the lateral ventricles and are commonly adherent to the septum pellucidum. A transcortical approach offers the best access to the tumour and potential for complete resection.

4. What are the typical microscopic characteristics of central neurocytoma?

Cells are uniform and round and have a benign appearance. They are WHO grade II.

5. What immunohistochemical markers are displayed in central neurocytoma?

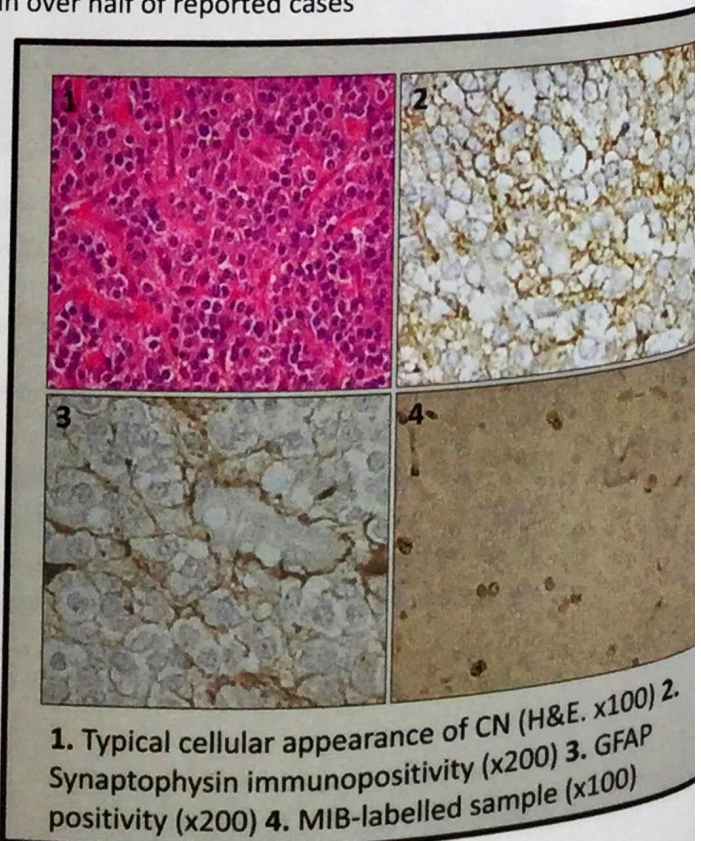
Synaptophysin: The most commonly displayed marker, almost all cases

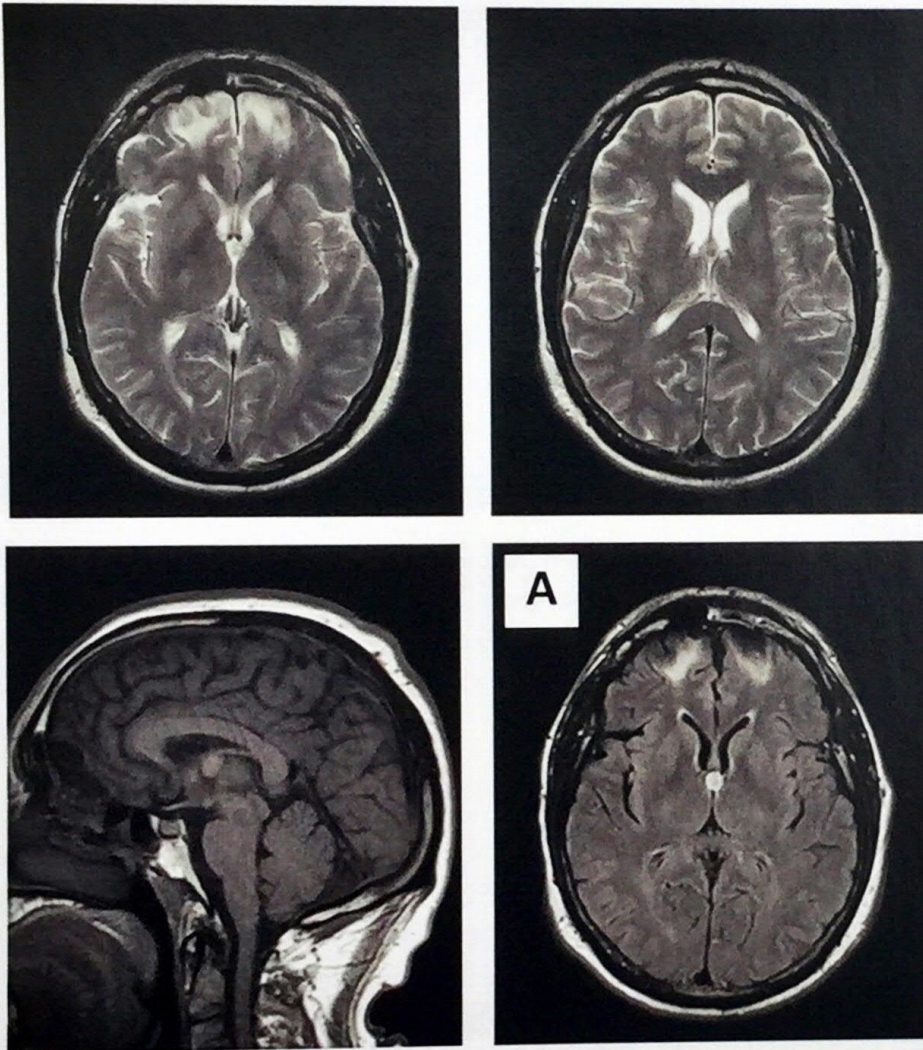
Neu N: Nuclei stain positively for Neu N in over half of reported cases

N52 & GFAP: Seen in half of cases

6. What is the MIB-1 index? What is the relevance in central neurocytoma?

MIB is a marker of cellular proliferation, based on Ki-67 antibodies. The MIB-1 index is the % of MIB positive nuclei in a given field of interest. In central neurocytoma, overall recurrence rates have been documented at 5% at 15 years. In cases where the MIB index is $>2\%$, recurrence rates can rise to as high as 60%.





This 41 year-old man is otherwise fit and well. He has a history of headaches for several months for which his GP has found no obvious cause.

Questions:

1. Describe the images shown; what is the most likely diagnosis?
2. What MRI sequence has been used in A, briefly explain this.
3. How would you explain the diagnosis and prognosis to the patient?
4. What are the management options for this patient?
5. If the patient presents in coma with radiological evidence of hydrocephalus, how would you manage this?

Answers:

1. Describe the images shown; what is the most likely diagnosis?

These are T2 axial (top row), T1 sagittal and FLAIR MRI images showing a lesion at the anterior aspect of the third ventricle. The most likely diagnosis is that of a colloid cyst. Less likely differentials include: astrocytoma, craniopharyngioma, papillary ependymoma and subependymoma.

2. What MRI sequence has been used in A, briefly explain this.

This is a FLAIR (fluid attenuated inversion recovery) sequence. Here, the 'pulse' phase of the MRI is an inversion recovery technique that can be configured to cancel out a particular substance, most commonly CSF in cranial imaging.

3. How would you explain the diagnosis and prognosis to the patient?

There is a single mucous-filled cyst in the brain. These are relatively rare (≈ 1 in 200 brain lesions). If cysts are small and do not grow (as many as 90% remain stable) they may never cause a problem. If they enlarge, they may obstruct CSF flow, causing hydrocephalus and other problems. There are reported cases of sudden death with these lesions although this is extremely rare.

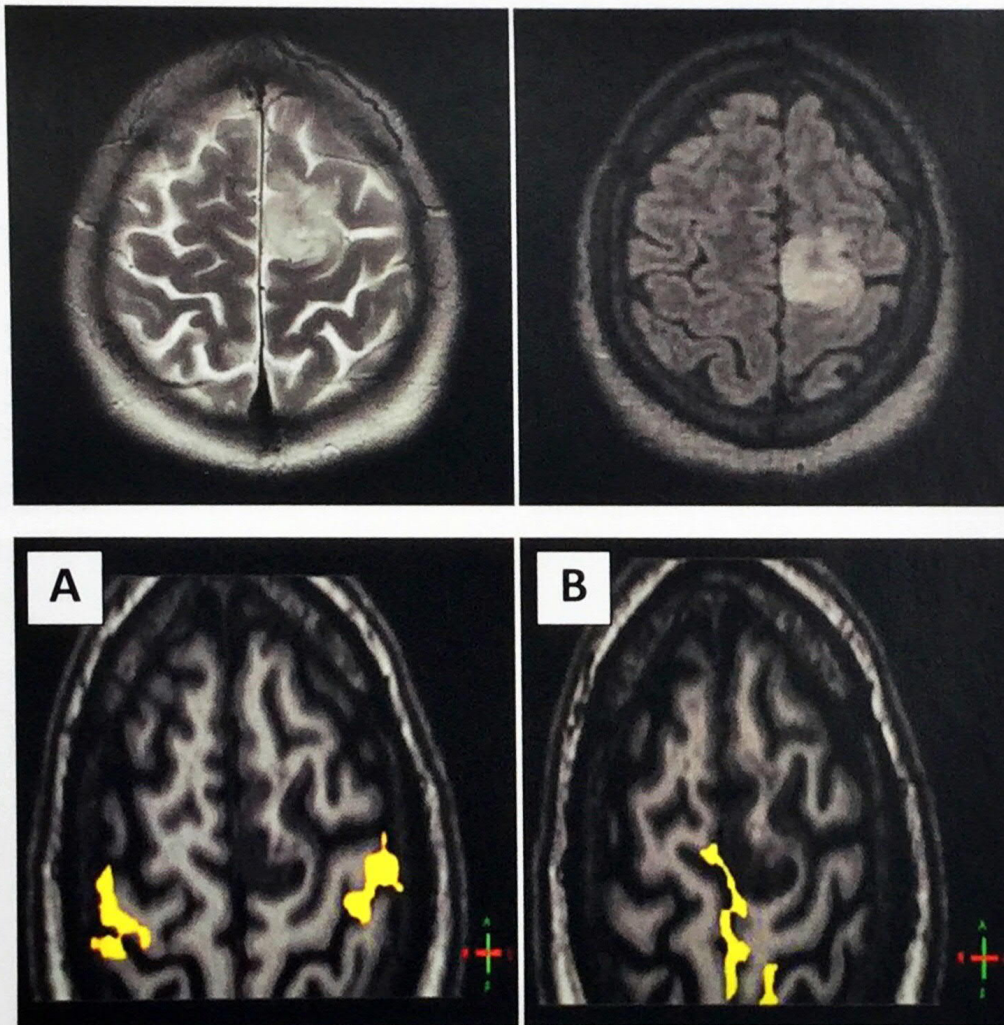
4. What are the management options for this patient?

Observational management may be appropriate for patients with asymptomatic lesions that do not grow, although even small lesions may produce symptoms. Surgical management may be *endoscopic* (fenestration or removal of the cyst $\pm$ septum pellucidum fenestration) or *microsurgical* excision (via craniotomy and transfrontal or transcallosal approach). Endoscopic management offers a minimally invasive approach and good visualisation of the cyst but haemostasis can be challenging, as can be complete excision. Microsurgery offers the best chance of complete excision but carries risks including seizures and memory loss (forniceal damage). *Stereotactic cyst aspiration* is possible although as the cyst is not excised, recurrence is more likely. *VP shunting* (either bilateral or unilateral with septal fenestration) may be appropriate for those who are not well enough to tolerate more aggressive management. *A transcallosal approach was performed for this patient in real life.*

A useful paper debating management; Hadjipanayis et al. Neurosurgery 2010; 67(1): 197-204

5. If the patient presents in coma with radiological evidence of hydrocephalus, how would you manage this?

Hydrocephalus must be dealt with promptly, either with bilateral EVDs or VP shunts. If a unilateral system is placed, attempts should be made to fenestrate the septum. Once the patient is stabilised, definitive management is as above. It is of course possible to attempt to perform this "in one sitting" (especially where endoscopy is used as placement will decompress the ventricles) if the expertise is available at the time of presentation.



This 35 year old presented with seizures. She has no prior medical history. She is initially treated medically but a second MRI (top row) shows subtle progression as compared with her initial scan (not shown). A functional MRI (fMRI) is performed prior to surgery (bottom row).

Questions:

1. Describe the appearance and location of this lesion.
2. How does fMRI work?
3. The fMRI images shown demonstrate finger tapping and toe curling. How is this useful?
4. How is an "awake craniotomy" performed?
5. What are the anatomical landmarks used for scalp blocks?